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Lithium Batteries Hot Paper

A Universal Principle of Lithium Bond Characteristics in Lithium Battery Electrolytes

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Abstract: Understanding the interactions between lithium (Li) and coordinated compounds is fundamental to elucidating the working mechanisms of Li batteries and informing the design principles of emerging battery materials. Li bonds, analogous to hydrogen bonds, play a central role in these interactions. However, the characteristics of Li bonds in working Li batteries are not fully explored due to the complexity of battery systems, hindering the advancement of Li bond theory in working batteries. In this work, a universal principle of Li bonds is established by combining theoretical calculations and machine learning techniques. Using organic electrolytes as modeling systems, the universal existence of Li bonds was first verified, particularly under high-salt-concentration conditions. A distinct characteristic of a coordination number of four was also demonstrated, distinguishing Li bonds from ionic bonds, consistent with electron localization region deviations around Li-ions. The influence of electrolyte components on Li bond characteristics was further revealed by ^{7}Li nuclear magnetic resonance (NMR) chemical shifts, and an ensemble machine learning model was constructed that quantitatively predicts ^{7}Li NMR chemical shifts with high accuracy. This work establishes a universal principle of Li bond characteristics in batteries, advancing fundamental Li chemistry, and guiding rational design for next-generation Li batteries.

Introduction

Lithium (Li) batteries (LBs) are essential for meeting the energy demands of contemporary applications, such as energy storage systems, electric vehicles, and the low-altitude economy.[1–4] Developing high-performance LBs with enhanced safety, high energy density, and low costs is highly required for the growing demands.[5–9] However, the lack of deep understanding of the working mechanism in LBs, such as ion transport kinetics and interfacial reaction mechanisms, seriously hinders the rational design of advanced battery materials to promote electrochemical performances.[10,11]

Unveiling the interaction between a Li center and coordinated components plays a central role in probing the working mechanism in LBs.[12,13] For instance, the Li-ion transport behaviors are highly dependent on the dynamics of Li-ion solvation structures, which are directly determined by the interaction between Li-ions and electrolyte components.[14–16] Beyond routine ionic bond theory, Li bonds, considered as analogs of hydrogen (H) bonds, were proposed to probe the bonding behaviors between Li and battery materials in the 2010s.[17–19] The introduction of Li bonds made great success in understanding the catalysis mechanism in sulfur cathodes and the lithiophilicity chemistry of Li metal anodes.[20–31] For instance, Li bonds have been proposed to reduce the dissolution of Li polysulfides in organic electrolyte in working Li–S batteries, thereby mitigating the shuttle of polysulfide intermediates and improving cycling stability.[32] Furthermore, during the Li deposition process on Li metal anode, Li bonds formed between a lithiophilic framework and Li-ions can guide uniform Li nucleation, and therefore deliver a dendrite-free composite anode.[33] These applications underscore the critical role of Li bonds in LBs and necessitate a deep understanding of Li bond characteristics.[34–37]

Despite similarities to H bonds, the metallic nature of Li affords Li bonds considerable ionic characteristics, unlike the partially covalent nature of H bonds.[38,39] The considerable ionic characteristics make Li bonds increasingly mysterious. For instance, a Li atom can be coordinated up to six atoms and can produce complicated structures, very different from those where only one H bond and one H covalent bond exist for one H atom. As a result, the fundamental principles of Li bond formation and its differences from the routine Li ionic bond (hereinafter referred to as ionic bond) are far from clear. Recently, Chen, Zhang, and coworkers proposed a distinct descriptor of a coordination number (CN) of 4 for distinguishing the Li bond and the ionic bond, taking 1,2-dimethoxyethane (DME) and bis(fluorosulfonyl)imide anion

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(FSI^-) systems as an example.[40] In the systems, the Li atom interacts with oxygen (O) atoms from electrolyte solvents or anions, forming the Li bond and ionic bond at low (CN 1–4) and high CN conditions (CN 5–6), respectively. Within the Li bond regime, an increasing CN induces a downfield shift of the ^{7}Li nuclear magnetic resonance (NMR) signal, indicating the electron localization effect of the Li bond. In contrast, for the ionic bond, the ^{7}Li peak shifts upfield due to the domination of electrostatic interactions. Besides, the difference between FSI^- and DME in regulating the ^{7}Li NMR chemical shift was also observed. Compared with DME, FSI^- generally induces a stronger shielding effect on a Li nucleus, forcing a down shift of the ^{7}Li chemical shift. The unignorable difference implies that the varied electrolyte components can afford Li bonds with great variety. However, the underlying universal principles have been rarely explored.

In this contribution, a general principle for Li bond characteristics is established by systematically probing the bonding behaviors and ^{7}Li NMR signals in various Li battery electrolytes through a combination of multi-scale simulations and machine learning methods. The ubiquitous existence of Li bonds in routine electrolytes was first confirmed, evidenced by a universal characteristic of a small CN no larger than 4. Besides, increasing salt concentration can obviously enhance the prevalence and average lifetime of Li bonds. Delving deep into the Li solvation structure, the universal boundary with a CN of 4 separate Li bond-dominated regions (CN 1–4) from ionic bond-dominated regions (CN 5–6), which is further correlated to the deviations in the electron localization region around Li-ions. Furthermore, a machine learning model was proposed to predict the ^{7}Li NMR chemical shift of various Li solvation structures, from which the highest occupied molecular orbital–lowest unoccupied molecular orbital (HOMO–LUMO) energy gap of corresponding electrolyte components was demonstrated to play an important role. This work advances fundamental Li chemistry and affords valuable theoretical and scientific guidance for the rational design of advanced Li battery electrolytes.

Results and Discussion

Dynamic Evolution of Li Bond

To verify the widespread existence of Li bonds in battery electrolytes and establish their universal characteristics, routine solvents and anions were selected, including DME, 1,3-dioxolane (DOL), ethylene carbonate (EC), dimethyl carbonate (DMC), NO_3^-, PF_6^-, bis(fluorosulfonyl)imide ($TFSI^-$), and FSI^-. Three kinds of salt concentrations, i.e., 1.05, 2.01, and 4.02 M, were considered, hereinafter referred to as low salt concentration (LC), medium salt concentration (MC), and high salt concentration (HC), respectively (Figure 1a–c). As a result, a total of 48 electrolyte models were built, of which the solvation structures were probed by molecular dynamics (MD) simulations.

Li^+ ions are coordinated with solvents or anions in working battery electrolytes, usually presenting a typical CN from 4 to 6. However, both coordinated solvents and anions can exchange with free solvents and anions that are not directly bound to a Li^+ ion, which is the origin of structural diffusion of Li^+ ions in electrolytes.[41] The exchange process can produce Li^+ ion structures with a CN smaller than 4. According to our previous reports, a CN of 4 can be adopted as the boundary to distinguish Li bonds from ionic bonds in electrolytes.[40] As a result, the ubiquitous existence of Li bonds across all the above 48 electrolyte systems was confirmed (Figure 1d,e). These Li bonds were observed to persist for continuous, finite durations. For instance, in the Li^+–DME–PF_6^- electrolyte system, Li bonds were found to persist for durations up to 0.8 ps, a lifetime that reflects their notable stability within the dynamic electrolyte environment.

The salt concentration was observed to exert a pronounced effect on the formation of Li bonds in electrolytes. As the concentration increases from LC to HC, both the proportion and average lifetime of Li bonds in all electrolyte systems steadily rise. Specifically, the Li bond proportions are approximately 3% with an average persistence of around 0.3 ps in LC electrolytes. In contrast, Li bond proportions often exceed 5% in HC electrolytes under HC conditions, with some systems even reaching over 40% such as Li^+–EC–$TFSI^-$. Simultaneously, the average lifetime of Li bonds typically surpasses 0.3 ps and extends beyond 1.0 ps in the aforementioned Li^+–EC–$TFSI^-$ system. The analysis of the solvation structures indicates that an increased number of anions populate the Li-ion solvation shell at higher concentrations, facilitating the assembly of large, negatively charged clusters (Figures S1–S4). These charged clusters act as "hard balls", impeding the penetration of external ions or molecules and thereby enhancing Li bond prevalence and average lifetime. Collectively, these findings demonstrate that HC electrolyte environments favor the formation and prolong the existence of Li bonds.

Besides salt concentration, the electrolyte components, especially the anion types, were also found to influence the formation of Li bonds. For instance, the NO_3^--containing electrolyte systems exhibit a high degree of anionic participation in the solvation shell, even at the LC electrolyte. The average number of coordinated anions in these NO_3^--containing electrolytes exceeds 3, a value substantially greater than that observed for systems with PF_6^- or $TFSI^-$, where the mean number of anions in the solvation structure is approximately 0 to 1 at the LC electrolyte (Figures S1–S4). Although not as obvious as anions, the role of solvents is not negligible. Electrolytes with DOL as the solvent were also found to promote high anion coordination, regardless of the anion type. At LC, the Li-ion solvation shell in DOL-containing electrolytes contains an average of 3 to 4 anions, in stark contrast to approximately 1 anion in electrolytes with other solvents (Figure S3). The observation can be attributed to the smaller molecular volume and lower steric hindrance of NO_3^- and DOL, allowing more anions to enter the first solvation shell of Li-ions. In contrast, solvents or anions with large volumes or strong steric hindrance (such as DMC, DME, and $TFSI^-$) limit the number of anions in the solvation structure.

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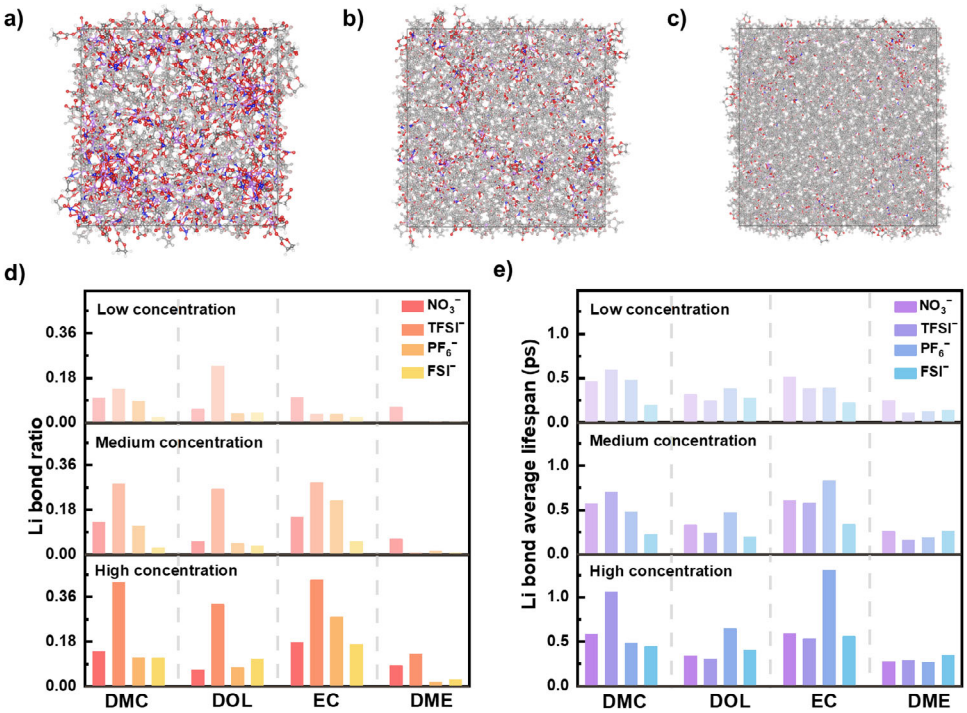


Figure 1. Li bonding analysis based on low-, medium-, and high-salt-concentration electrolytes. Three electrolytes are denoted as LC, MC, and HC, respectively, including various solvents (DMC, DOL, EC, and DME) and anions (NO_3^-, FSI^-, PF_6^-, and $TFSI^-$). Snapshots of MD simulations on a) HC, b) MC, and c) LC electrolytes, taking Li^+–EC–NO_3^- electrolyte system as an example. Atomic color: H, white; Li, violet; C, gray; N, blue; and O, red. d) Li bond ratio and e) Li bond average lifespan (ps) with different solvents and anions across the three concentration regimes.

Characteristics of Li Bonds

To unveil the underlying mechanism of electrolyte components in regulating Li bond characteristics, DFT calculations were further conducted to probe all possible Li-ion solvation structures with a CN from 1 to 6. Li-ions are coordinated with O or F atoms from solvents (DME, DOL, EC, and DMC) or anions (NO_3^-, PF_6^-, $TFSI^-$, and FSI^-), producing 157 structures in total (Figures S5–S22). Inspired by previous studies on H bond characterization, ^{7}Li NMR spectroscopy was also employed as the primary tool to explore the Li bond chemistry.[40–42]

The ^{7}Li NMR chemical shifts were first correlated to the CN, and a "volcano" plot was observed (Figure 2a), which is consistent with the previous report in the DME and FSI^- system.[40] Specifically, the ^{7}Li chemical shift gradually moves downfield as CN increases from 1 to 4, and then turns upfield when the CN further increases to 6. These two opposing trends correspond to distinct bonding regimes: the low-coordination region (CN 1–4) is characteristic of Li bonds, while the high-coordination region (CN 5–6) is dominated by ionic bonds. The volcano trend reflects a fundamental change in the electronic environment of the Li nucleus. The initial downfield shift indicates progressive deshielding, as the formation of Li bonds with increasing electron localization effect. In contrast, the subsequent upfield shift signifies enhanced shielding due to the domination of electrostatic interactions. Collectively, these divergent trends and their underlying chemical origins establish the turning point at CN = 4 as a clear demarcation between the Li bond and ionic bond regimes.

Li bonds deliver different bonding characteristics from ionic bonds (Table S1). On average, Li bonds exhibit a shorter bond length of approximately 1.9 Å and a weaker bond energy of about −3.7 eV. This contrasts sharply with ionic bonds, which display a mean bond length of approximately 2.1 Å and a larger bond energy of around −4.9 eV (Figures S23 and S24). In this case, the "bond" here, which considers all Li–O bonds to accord with the NMR situation, and the traditional bond between only two atoms has a twist on the meaning. Notably, these intrinsic properties are largely independent of the specific solvent and anion species present across all studied electrolytes, with the mean bond length near 2.0 Å and energy remaining consistently −4.1 eV, respectively. This consistency across disparate chemical environments not only highlights the universal nature of these two bond types but also further validates the clear boundary between them.

To understand the underlying mechanism of Li bond characteristics, the electronic structures of Li-ions were further analyzed. The ^{7}Li NMR chemical shifts are primarily influenced by diamagnetic and paramagnetic shielding effects from the electronic environment around the nucleus, in which the electron density and its spatial distribution are key factors.[43,44] According to the natural bond orbital (NBO) analyses, the natural population analysis (NPA) charge on Li-ions gradually reduces as the CN increases (Figure S25). Intuitively, an increased electron density should enhance

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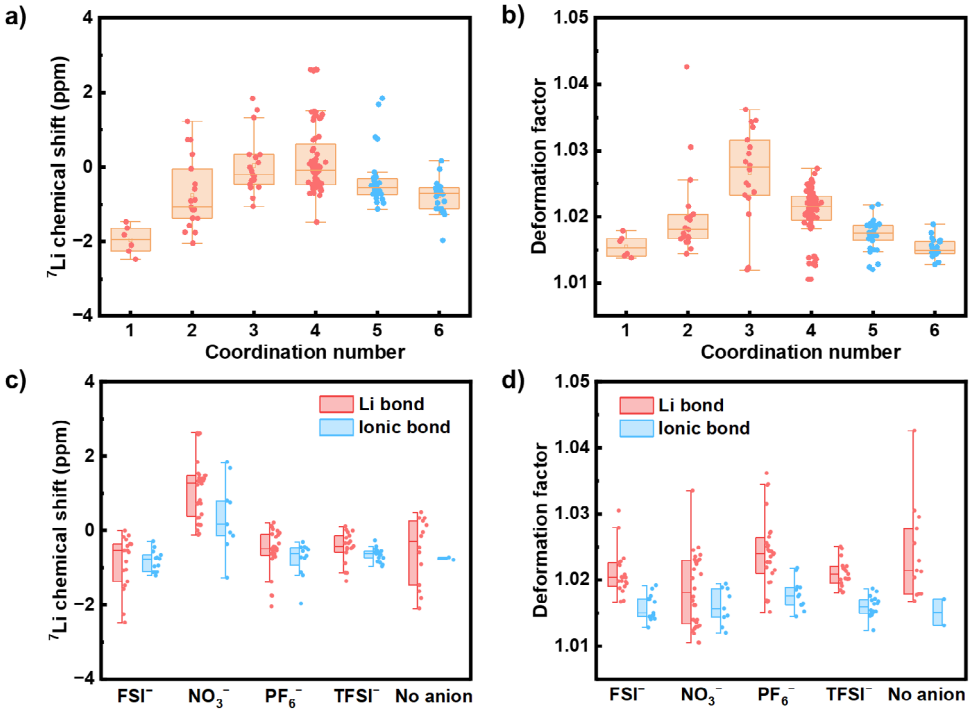


Figure 2. Comparison between Li bond and Li ionic bond. a) Change of ^{7}Li chemical shift as Li coordination number. b) Change of ELF deformation factor as Li coordination number. An isovalue of 0.75 was adopted. c) Comparison of ^{7}Li chemical shifts for different anions ($TFSI^-$, FSI^-, PF_6^-, NO_3^-, and no anion coordinated). d) Comparison of ELF deformation factors for different anions ($TFSI^-$, FSI^-, PF_6^-, NO_3^-, and no anion coordinated).

shielding effects and lead to an upfield shift in chemical shifts. However, a downfield ^{7}Li chemical shift and a clear deshielding effect are observed in the Li bond region (Figure 2a). Therefore, the NPA charge can only consider the diamagnetic shielding effects.

To comprehensively consider both the diamagnetic and paramagnetic shielding effects, a deformation factor (ϕ) derived from the electron localization function (ELF) was introduced to characterize the spatial distribution and localization degree of electrons around Li-ions.[45] ELF reflects the localization characteristics of electrons in three-dimensional space, and the deformation factor ϕ quantifies the deviation of ELF from ideal spherical symmetry (more details can be found in DFT in the Supporting Information). For bare Li-ions, the deformation factor ϕ (1.003) is very close to 1, indicating a spherically symmetric electronic environment. As CN increases, ϕ first increases from 1.015 to 1.025 and then drops back to 1.015 (Figure 2b). The trend highly coincides with the "volcano" behavior of ^{7}Li chemical shifts. Specifically, when CN increases from 1 to 4, the variation of ϕ reflects the deviation of the electronic environment from spherical symmetry, leading to weakened shielding effects and a downfield shift in chemical shifts. When CN increases from 4 to 6, the reduction in ϕ indicates a tendency toward more symmetric electronic environments, enhancing shielding effects and causing an upfield shift in chemical shifts (Figure 2a,b). Consequently, the deformation factor ϕ effectively captures the overall changes in the electronic environment around Li-ions and affords a reasonable explanation for the non-monotonic trend of ^{7}Li chemical shifts.

Similar to the regulation on solvation structures, electrolyte components can determine the ^{7}Li NMR chemical shifts and ϕ. Solvents have a minor effect on the NMR characteristics of Li bonds in electrolytes (Figures 2d and S26). No obvious difference in ^{7}Li NMR chemical shifts and ϕ was observed for DME, DOL, EC, and DMC (Figure S27). However, NO_3^- induces a stronger deshielding effect and results in a larger chemical shift compared with PF_6^-, $TFSI^-$, and FSI^- (Figure 2c). Specifically, the average chemical shift within the Li bond range is around 1.1 and −0.6 ppm for NO_3^- and the other anions, respectively. Similarly, the chemical shift within the ionic bond region is 0.4 and −0.7 ppm for NO_3^- and the other anions, respectively. The pronounced upfield shift for NO_3^- is attributed to its stronger interaction with a Li-ion. This interaction profoundly alters the local electronic environment around a Li-ion, rather than the changes in geometrical structure, especially considering that the species of solvents and anions have little impact on bond energy and bond length (Figure S24). Different from chemical shifts, no obvious variation was observed in ϕ among different anions (Figures 2d and S26). Therefore, ϕ primarily reflects the influence of CN on the electronic environment, rather than the nature of specific coordinating species.

Data-Driven Analysis of Li Bond Characteristics

In order to understand the underlying reasons why different electrolyte components have varied influences on Li bond characteristics, an interpretable machine learning

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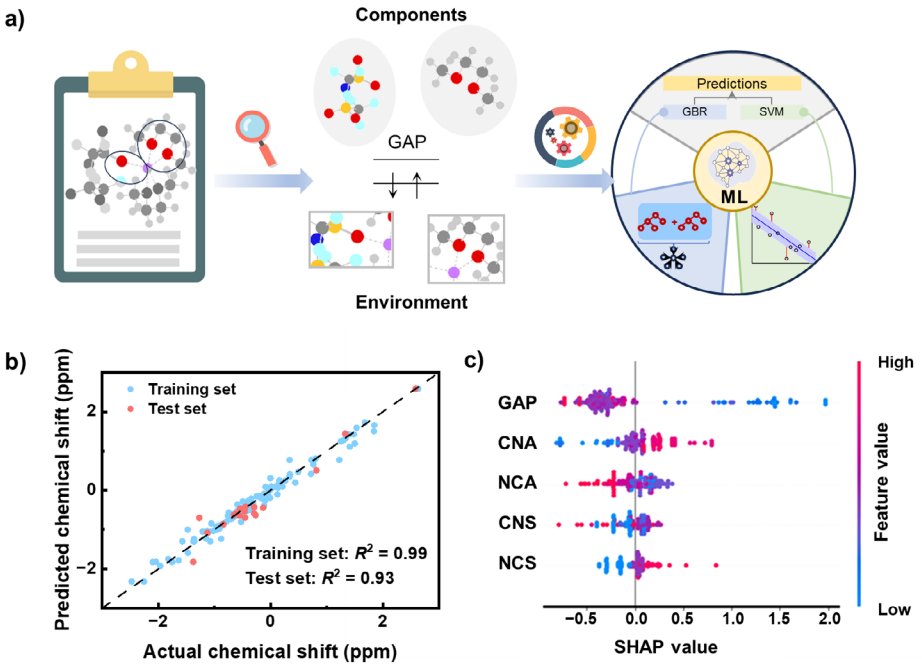


Figure 3. Establishment and analysis of machine learning model based on DFT calculation. a) Schematic introduction of machine learning workflow. b) Parity plots between the actual and predicted ^{7}Li chemical shift. c) Interpretable analysis by the SHAP value.

model was further proposed to predict ^{7}Li NMR chemical shift (Figure 3a). The machine learning model employs the following five descriptors: the coordination number of anions (CNA) and solvent molecules (CNS); the number of coordinating anions (NCA) and solvent molecules (NCS); and a weighted-average HOMO–LUMO energy gap (GAP) of the coordinating species, with the average weighted by the respective numbers of coordinating anions and solvent molecules (Table S2). Especially, the GAP is related to the energy difference between the ground state and the first excited state, which is correlated to the NMR shielding tensor within the DFT framework (Figure S28).[46,47] A Pearson correlation coefficient (PCC) analysis confirmed that no strong multicollinearity exists among these descriptors, validating their suitability for the model (Figure S29). Both gradient boosting regression (GBR) and support vector regression (SVR) algorithms were employed for model training, and a final ensemble learning model was constructed through a voting integration method (Figure S30). After a hyperparameter optimization, the ensemble model achieved a coefficient of determination of 0.93 on the test set for predicting the ^{7}Li NMR chemical shifts of solvation structures (Figure 3b).

Shapley additive explanations (SHAP) analyses were subsequently employed to thoroughly explore the contribution and influence of each input feature on the model's predictions. As expected, the GAP delivers the most important effects (Figure 3c). According to the DFT framework, a smaller energy difference between the ground state and excited states leads to stronger paramagnetic shielding effects and a downfield shift in ^{7}Li chemical shifts. Additionally, the coordination of anions renders a much greater impact on chemical shifts than that of solvents, underscoring the dominant role of anions in Li bond behavior. For instance, NO_3^-, due to its small HOMO–LUMO gap, renders a notable downfield shift in ^{7}Li chemical shift in its coordination structures.

The machine learning model was subsequently employed to predict the ^{7}Li chemical shifts of the 48 practical electrolyte systems, which feature diverse compositions and concentrations, by averaging the values derived from individual solvation structures (Figure 4a). As a result, the predicted chemical shifts of all solvation structures mainly fall within the range of −1 to 0 ppm with high frequency (Figures 4b and S31). Upon averaging, the final predicted chemical shifts for the different electrolyte systems converge within a narrower range of −0.2 to −0.6 ppm (Figure 4c). The salt concentration was identified as a primary factor governing the ^{7}Li chemical shift, corroborating the MD and DFT analyses. Specifically, as the concentration increases from LC through MC to HC for the DME and FSI^- system, the model predicts a progressive upfield shift from −0.46 to −0.52 and finally −0.58 ppm. This result exhibits strong quantitative agreement with previously calculated shifts (from −0.49 to −0.53 and −0.61 ppm) and aligns with the qualitative trend observed experimentally.[40]

As demonstrated above, the machine learning model not only possesses excellent robustness in predicting Li bond characteristics across diverse electrolyte compositions, but also serves as a powerful computational tool for the high-throughput rational design of novel electrolyte systems. On the one hand, the model affords a significant advantage by directly linking atomistic solvation structures to experimental spectroscopy, which helps to unveil electrolyte chemistry. On the other hand, the existence of Li bonds is supposed to accelerate Li-ion transport behaviors due to the weaker binding energy of Li bonds compared with ionic bonds. High-throughput screening can be conducted based on the machine

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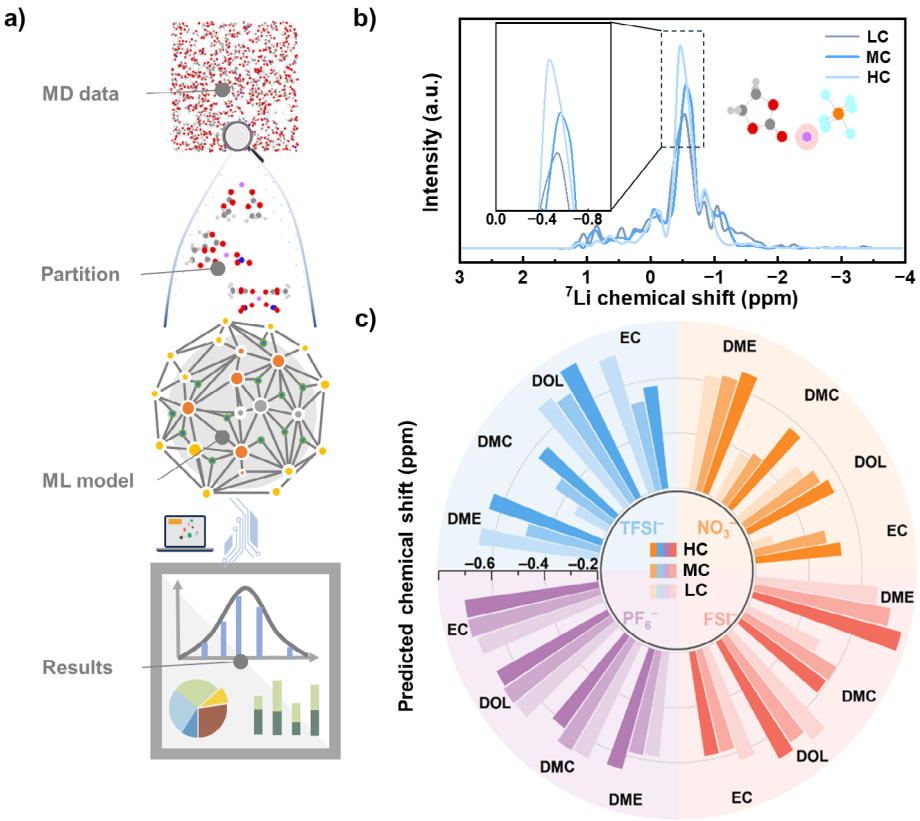


Figure 4. Predictions made by a machine learning model based on MD simulation results. a) Schematic introduction of the model training–predicting workflow. b) Distribution of predicted ^{7}Li chemical shift for LC, MC, and HC electrolytes. c) Prediction of average ^{7}Li chemical shift for various solvents (EC, DOL, DMC, and DME) and anions (TFSI$^-$, FSI$^-$, PF_6^-, and NO_3^-) in LC, MC, and HC electrolytes.

learning model to rationally design advanced electrolytes with tailored solvation structure and promising ionic diffusivity.

Conclusions

This contribution establishes a universal principle of Li bond characteristics in Li battery electrolytes by combining multi-scale simulations and machine learning models. The wide existence of Li bonds in various electrolytes was first confirmed, in which the salt concentration has a significant influence on the percentage and lifetime of Li bonds. HC electrolyte notably promotes direct anion–Li^+ coordination within solvation structures, indirectly enhancing Li bond stability, leading to a prevalence often exceeding 5% and an average lifetime typically above 0.3 ps. A universal principle of a critical CN of 4 was demonstrated across all the investigated electrolyte systems that Li bonds and ionic bonds appear in the CN region of 1–4 and 5–6, respectively. The distinction arises from deviations of the electron localization function around Li-ions due to coordination changes, leading to contrasting ^{7}Li NMR chemical shift trends between the two regions. The influence of electrolyte components on Li bond characteristics was further analyzed by both direct visualization and interpretable machine learning models. Compared with solvents, anions have a more obvious influence on ^{7}Li NMR chemical shift. The weighted-average HOMO–LUMO energy gap of electrolyte components was confirmed to be the most important descriptor for predicting ^{7}Li NMR chemical shifts. These findings advance fundamental Li bond chemistry and afford fruitful theoretical guidance for the rational design of advanced electrolyte materials for high-performance Li batteries.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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